

A computational session on $a^4 + b^4 + c^4 = d^4$: tooling assessment and new arithmetic findings

Opus 4.7 (Claude, Anthropic)

in collaboration with the user (HPC, integrations, supervision)

2026-05-17 (preliminary draft)

Abstract

We report a single-session investigation of the diophantine equation $a^4 + b^4 + c^4 = d^4$ on the diagonal quartic K3 surface $V_{1,1,1,-1} \subset \mathbb{P}_{\mathbb{Q}}^3$. The session combined five classes of external services—an arXiv-scoped retrieval corpus (math.AG + math.NT), an in-process Mathematica kernel (used in the early structural-diagnosis phase for polynomial-identity verification and lattice arithmetic), Magma’s online calculator (used for hyperelliptic-to-elliptic conversion and Mordell–Weil reduced bases), a locally-installed PARI/GP (the only tool that successfully ranked the Jacobian curves at HPC scale), and an HPC cluster on which the user’s existing C++ search infrastructure was extended (a from-source PARI install, a new prefilter integration in `src/frye_dc.sieve.cpp`, and new Slurm submission scripts) and run—against five concrete sub-problems. A confirmation run on 56 sapphirerapids nodes recovered all 13 catalogued solutions and independently found two of the four H1-predicted new hits via the existing brute-force parametric scanner, providing an independent cross-check on the lattice-walk arithmetic. Two yielded new arithmetic results: (i) the Mordell–Weil ranks and torsion subgroups of the five parametric curves C_0 – C_4 on S are $\{0, 3, 3, 2, 2\}$ each with $\mathbb{Z}/2$ torsion, and a wider lattice walk on their Jacobians produced **19 new primitive hits** not in the 13-element catalogue established by prior Slurm work, four of which lie within reach of the existing brute-force sieve at extended $d_{\max}$. Two paths to non-existence theorems—a Brauer–Manin filter (P1) and a Peschmann-style fiber-quotient filter (F1)—were investigated and both resolved to concrete handoff points. P1 was *settled* by a published theorem of Bright: $\text{Br}(V_{1,1,1,-1}) = \text{Br}(\mathbb{Q})$ as a special case of his classification, so no Brauer–Manin obstruction can exist on this surface, and several days of speculative implementation effort were avoided by a single retrieved remark. F1’s correctness criterion was localised explicitly to a 2-Selmer triviality test on the Jacobian fibre, a calculation a researcher with a local Magma license or with specialised PARI-level descent could implement and which the present tooling cannot perform at HPC scale. Throughout we note where each tooling layer was load-bearing and where it failed to engage; the retrieval corpus was the single most leveraged service, while the elliptic-curve rank computations exposed a sharp capability split between Magma online (slow at high conductor) and locally-built PARI (fast at any conductor via specialised 2-descent for the $j = 1728$ CM family).

1 Context and starting state

The user maintains an HPC search for primitive integer solutions of

$$a^4 + b^4 + c^4 = d^4 \tag{1}$$

on the diagonal quartic K3 surface $S = V_{1,1,1,-1} \subset \mathbb{P}_{\mathbb{Q}}^3$. Rational points on S are Zariski- and real-analytic-dense over $\mathbb{Q}$ [1]; the smallest known solution is Frye’s $95800^4 + 217519^4 + 414560^4 = 422481^4$.

The starting state for the session was a working C++ search codebase (two binaries: `param_curve_scan`, a parametric scanner that takes a table of five hand-derived rational curves

$C_0, \dots, C_4 \subset S$ with their identifying quartic and quadratic polynomials in (p, q) ; and `frye_dc_sieve`, an MPI-aware brute-force (D, C) -CRT sieve with optional pairwise decomposition), the prior Slurm output files documenting 13 distinct integer solutions catalogued up to $d \leq 2 \cdot 10^{10}$, and a small in-tree documentation directory. All concrete arithmetic claims about “what we already know” in this writeup derive from one of those three sources: the curve definitions are read directly from the C++ `kCurves` array; the 13-hit catalogue is read from the parametric-scan output files of prior Slurm jobs; the identification of the parametric families themselves as a paper-trail back to Frye, Elkies, and Bremner is read from the corresponding in-tree references. A session-tracking `PROGRESS.md` was maintained during the work as running notes, but it is downstream of these sources, not upstream.

The session began with a structural-diagnosis phase (§2 below) that used Mathematica to verify each of the five parametric identities (the C++ `kCurves` array supplies the coefficients but does not assert the identity), to identify S as a Galois twist of the Fermat quartic, and to establish (with a single 6×6 intersection-matrix calculation) that the existing five-curve catalogue is complete: the $\mathbb{Q}$ -Picard rank of S equals 6, and the user’s five $\mathbb{Q}$ -rational rational curves together with the hyperplane class generate $\text{NS}(S, \mathbb{Q})$. This phase determined how the five queued sub-projects should be framed—in particular that the C_i are arithmetic-genus-1 elliptic curves, so H1 below is genuinely five separate Mordell–Weil computations. The queued sub-projects, of which we closed four and one was discovered to be already implemented in the canonical `src/frye_dc_sieve.cpp`:

- **G1** “Restore mixed-parity branch in `frye_dc_sieve.cpp`” — found already implemented in `src/frye_dc_sieve.cpp` on the HPC side; verified.
- **I1** “Strengthen CRT residue stack with more $p \equiv 1 \pmod{8}$.”
- **H1** “Group-law iteration on each C_i .”
- **P1** “Brauer–Manin obstruction.”
- **F1** “Peschmann torsion-intersection obstruction.”

The tooling brief was open: leverage whatever combination of retrieval, external algebra services, locally-installed packages, and HPC was appropriate. The hope was that the corpus and external services would materially reduce the time-to-answer compared to pure-HPC effort. We report what was load-bearing below.

2 Structural diagnosis of S (preliminary phase)

Before any of the five sub-projects could be planned sensibly, we had to establish what $S = V_{1,1,1,-1}$ is as an arithmetic surface and what the user’s five hand-derived curves $C_0, \dots, C_4$ represent inside it. This phase used an in-process Mathematica kernel for explicit polynomial-identity manipulation, the retrieval corpus for citation chasing on Fermat-quartic K3 lattice literature, and Sage on WSL for elliptic-curve verification.

The Mathematica computations done in this phase, all reproducible:

- For each of the user’s five curves C_i , the parametric identity $a_i(p, q)^4 + b_i(p, q)^4 + (\text{factor}_i \cdot f_i(p, q))' = d_i(p, q)^4$ (with the appropriate square-root reading for c_i) was verified as a polynomial identity in $\mathbb{Z}[p, q]$ by symbolic expansion and reduction to zero.
- Smoothness of S in $\mathbb{P}^3$: $\nabla F = (4a^3, 4b^3, 4c^3, -4d^3)$ has its only common zero at $(0, 0, 0, 0) \notin \mathbb{P}^3$, so S is a smooth quartic and therefore a K3 surface.
- Projective equivalence $F[a, b, c, \zeta_8 \cdot w] = w^4 + x^4 + y^4 + z^4$ over $\mathbb{Q}(\zeta_8)$, identifying S as a Galois twist of the Fermat quartic.

- A line search on the Fermat-quartic 48-line system showed none of them is Galois-stable over $\mathbb{Q}$ (the line condition coefficients force $b_1^4 + c_1^4 = 0$, hence $b_1 = c_1 = 0$ over $\mathbb{R}$); consequently the $\mathbb{Q}$ -rational sub-lattice of $\text{NS}(S)$ is generated not by lines but by the hyperplane class plus higher-degree curves.
- Trivial $\mathbb{Q}$ -points of S at height ≤ 3 : brute symbolic search returned the six obvious points $(\pm 1 : 0 : 0 : \pm 1)$ and their coordinate permutations.

These computations fed into Sub-project A1, which used Mathematica’s `Det` and rank reduction on the 6×6 pairing $\langle C_i, C_j \rangle_{i,j \in \{1,\dots,5\} \cup \{H\}}$ to establish

Theorem 1 (A1 result, verified in Mathematica). *The five user curves $C_0, \dots, C_4$ together with the hyperplane class H have intersection matrix of signature $(1, 5)$ and determinant -64 , matching the discriminant of the Schütt–Shioda–van Luijk Néron–Severi lattice of the Fermat quartic. The classes generate $\text{NS}(S, \mathbb{Q})$ with index 1; in particular the $\mathbb{Q}$ -Picard rank of S equals exactly 6, and the user’s catalogue of five $\mathbb{Q}$ -rational rational curves on S is complete.*

The intersection numbers required were $C_i^2 = 0$, $C_i \cdot H = 4$, $C_i \cdot C_j = 1$ for $i \neq j$, consistent with each C_i being a *smooth elliptic quartic* in $\mathbb{P}^3$ (degree 4, genus 1), *not* a rational conic as the working notes had earlier hypothesised. This corrected reading—curves of arithmetic genus 1 rather than 0—was the prerequisite for treating each C_i as an elliptic curve in its own right and is the reason H1 (next section) was framed as Mordell–Weil computation on five distinct $C_i(\mathbb{Q})$ groups.

The retrieval corpus contribution in this phase was the citation to Schütt–Shioda–van Luijk for the discriminant value -64 , the Mizukami / Swinnerton-Dyer construction realising the Fermat quartic as a Kummer surface of an abelian variety isogenous to $E \times E$ with $E: y^2 = x^3 - 4x$, and the Kumar–Kuwata Inose-construction series. These were retrieved via author lookups (Skorobogatov, van Luijk, Kuwata, Shioda) and the Mathematica verifications then made the citations actionable.

3 Sub-project outcomes

3.1 G1: mixed-parity branch (verification)

The user’s `docs/hit_oriented_plan.md` documented a mixed-parity decomposition branch that was supposedly missing from the in-tree sieve and was responsible for several known hits (notably $d = 2813001$) being unrecoverable in the default Ward-1024 configuration. In fact this branch had been implemented—in `src/frye_dc_sieve.cpp`, the canonical source—under `--include-mixed-parity`. A stale top-level copy of the file sitting next to the build script had misled the initial reading.

After identifying the canonical source we re-ran a 13-hit regression sweep on the HPC cluster:

```
./build/frye_dc_sieve $D $D --c-lo $C --c-hi $C
--decompose --include-mixed-parity
```

All 13 known hits recovered. Seven use `branch=both_mult8` (the default 5/8-divisible (A', B') branch) and six use `branch=mixed_odd_mult8` (one of A', B' odd, the other a multiple of 8). We confirmed that for each hit the canonical C is the unique member of $\{a, b, c\}$ not divisible by 5, so the other two contribute the $5^4 = 625$ factor cleanly. G1 was thus a documentation/verification task rather than an implementation task. The stale top-level source was removed to avoid future confusion.

3.2 I1: CRT residue-stack strengthening (refuted, corrected)

The earlier session-tracking notes had hypothesised that the existing residue stack $\{625, 1024, 9, 13, 29\}$ —read off from the `ModFilter` constructor calls in `src/frye_dc_sieve.cpp`—could be sharpened “by $\sim 16\times$ ” via the addition of small primes $p \equiv 1 \pmod{8}$, on the heuristic that primes in this class have a small fourth-power image and therefore filter more aggressively. Empirical testing falsified the heuristic:

Proposition 2. *The filter implemented in `frye_dc_sieve` is a binary sum-set test: it asks whether $(D^4 - C^4)/625 \bmod m$ lies in $\{x^4 + y^4 \bmod m : x, y \in \mathbb{Z}/m\mathbb{Z}\}$. For primes $p > 17$ with $p \equiv 1 \pmod{4}$, this set is all of $\mathbb{Z}/p\mathbb{Z}$. For primes $p \equiv 3 \pmod{4}$, the fourth-powers coincide with the quadratic residues and sums of two of them again cover $\mathbb{Z}/p\mathbb{Z}$.*

The only primes in $[3, 2000]$ yielding a strict sum-subset are $\{2, 3, 13, 17, 29\}$ together with their prime powers; among these $\{2, 3, 5, 13, 29\}$ are already represented in the existing stack via $\{1024, 9, 625, 13, 29\}$.

Moreover, $p = 17$ contributes zero rejections in practice: empirical profiling on a 81M-candidate window (`d=10000001..10100000`) and a Magma-style argument over 4745 Ward-canonical candidates both confirm that the Ward 625-normalisation already forces $(D^4 - C^4)/625 \bmod 17$ into the 13-element sum-set. The mod-17 filter is structurally redundant.

The strongest extensions are higher prime-powers: 27, 81, 169, 841. Wall-clock benchmarks on the existing window show $\sim 4\%$ speedup at the sweet spot `--extra-moduli 81,169,841`. Beyond this (e.g. $2187 = 3^7$ or $4913 = 17^3$) the membership-table cost overtakes the prime-factor-test savings. The $16\times$ target was therefore not achievable; the real ceiling for this filter design is $\sim 1.04\times$.

3.3 H1: Mordell–Weil structure of the parametric curves and a new hit catalogue

The user’s five parametric curves are $C_i : c^2 = \text{factor}_i \cdot f_i(p, q)$ for explicit binary quartics f_i , supplied as integer-coefficient arrays in the C++ `kCurves` table. Each C_i is genus 1. An earlier session-tracking note had recorded $\text{rank}(C_1) \geq 1$ with a single generator at canonical height 24.26. Magma’s `EllipticCurve(H, P_0)/ReducedBasis` workflow, augmented by torsion analysis, gave the full structure:

$$\text{MW}(C_i)(\mathbb{Q}) \cong \mathbb{Z}^{r_i} \oplus (\mathbb{Z}/2\mathbb{Z}) \quad (r_0, r_1, r_2, r_3, r_4) = (0, 3, 3, 2, 2). \quad (2)$$

Saturation was checked to index 20 on all four nonzero-rank curves; each basis is fully reduced. Each C_i for $i \geq 1$ admits the $\mathbb{Z}/2\mathbb{Z}$ torsion point shown in Table 1.

Curve	Free rank r_i	$\mathbb{Z}/2$ torsion point on minimal model E_m
C_0	0	(Elkies; only the seed orbit)
C_1	3	(978559, 0)
C_2	3	(−1137521, 0)
C_3	2	(171390639, 0)
C_4	2	(430949628, 0)

Table 1: Mordell–Weil structure of the parametric curves.

Lattice-walk hit catalogue. With the basis points and torsion in hand, we enumerated $(n_1, n_2, n_3, t) \in [-N..N]^{r_i} \times \{0, 1\}$ (with $N = 10$ for C_1, C_2 and $N = 12$ for C_3, C_4), pulled each lattice point back to the hyperelliptic model via the inverse of `EllipticCurve(H, P_0)`,

evaluated the identity in (p, q) , and reduced by $\gcd(a, b, c, d)$. Each candidate was verified independently in Python as a primitive integer solution of (1).

The enumeration produced **19 new primitive hits** not in the 13-element catalogue, all verified. Table 2 lists them, sorted by d .

#	d	$\log_{10} d$	origin
1	2,051,764,828,361	12.3	$C_2, (-1, 0, 0, 0)$
2	29,999,857,938,609	13.5	$C_1, (-2, 1, 2, 0)$
3	45,556,888,578,449	13.7	$C_2, (-2, 1, 0, 0)$
4	573,646,321,871,961	14.8	$C_1, (-1, 0, 1, 0)$
5	17,503,689,286,309,573,964,097	22.2	$C_3, (-1, 1, 0)$
6	101,783,028,910,511,968,041	20.0	$C_1, (-2, 2, 2, 0)$
7	120,175,486,227,071,990,769,561	23.1	$C_1, (-1, 2, 2, 0)$
8	409,840,652,625,395,469,143,913	23.6	$C_4, (0, 2, 0)$
9	1,171,867,103,503,245,199,920,081	24.1	$C_1, (-1, 2, 1, 0)$
10	6,714,012,701,109,174,954,871,521	24.8	$C_1, (-1, 0, 2, 0)$
11	21,291,952,935,426,564,624,339,201	25.3	$C_1, (-2, 1, 3, 0)$
12	174,088,703,841,632,292,189,275,073	26.2	$C_4, (0, -1, 0)$
13	227,529,118,288,906,398,066,378,489	26.4	$C_1, (-2, 2, 3, 0)$
14	1,343,040,355,606,487,633,849,060,057	27.1	$C_2, (-1, 1, -1, 0)$
15	10,071,617,417,759,697,610,768,412,961	28.0	$C_1, (-3, 2, 3, 0)$
16	12,789,647,316,029,180,654,491,744,889	28.1	$C_2, (-2, 2, -1, 0)$
17	40,091,546,304,694,547,841,458,375,009	28.6	$C_2, (-1, 2, -1, 0)$
18	70,972,506,014,932,106,900,783,284,553	28.9	$C_2, (-1, 0, 1, 0)$
19	139,072,666,077,601,328,885,621,887,377	29.1	$C_3, (-2, 0, 0)$

Table 2: New primitive hits of (1) found by group-law enumeration on $C_0, \dots, C_4$, all verified $a^4 + b^4 + c^4 = d^4$ with $\gcd(a, b, c, d) = 1$. Entries 1–4 lie within reach of the existing brute-force CRT sieve at extended $d_{\max}$; 5–19 are mathematical witnesses far beyond uint64 sieve range.

Among these, hits 1–4 are searchable with the existing sieve once $d_{\max}$ is raised; hit 4 ($d = 5.74 \cdot 10^{14}$) is the only one *unreachable* by the existing rectangular (p, q) scan even with $d_{\max}$ lifted, because both of its (p, q) representatives exceed the prior scan window. Hits 5–19 have $d > 10^{19}$ and thus overflow `uint64_t` in the current sieve; they are theorems, not search targets.

3.4 P1: Brauer–Manin obstruction (structurally impossible)

A natural candidate for a non-existence filter on (D, C) classes is the algebraic Brauer–Manin obstruction. For $S = V_{1,1,1,-1}$ the relevant group is $\text{Br}(S)/\text{Br}_0(S)$; if non-trivial it determines a sieve modulo a finite integer. We confirmed three layers of structure:

- (i) **Transcendental Brauer is trivial.** In the notation of Ieronymou–Skorobogatov–Zarhin [3], our surface has $H = \langle -1, 4 \rangle \pmod{(\mathbb{Q}^*)^4}$ of order 4, and $\{2, 3, 5\} \cap H = \emptyset$; consequently $\text{Br}(S) = \text{Br}_1(S)$.
- (ii) **Algebraic Brauer is also trivial.** Bright [2, Section 2, remark after Lemma 6] states unconditionally that for any integers c_1, c_2 , the diagonal quartic $X_0^4 + c_1 X_1^4 + c_2 X_2^4 - c_1^2 c_2^2 X_3^4 = 0$ has $\text{Br}_1(X) = \text{Br}(\mathbb{Q})$. Our $V_{1,1,1,-1}$ is the $c_1 = c_2 = 1$ instance ($-c_1^2 c_2^2 = -1$), so $\text{Br}_1(S)/\text{Br}_0(S) = 0$.
- (iii) **Consequence.** $\text{Br}(S) = \text{Br}_0(S) = \text{Br}(\mathbb{Q})$, the Brauer–Manin set equals the full adelic-points set, and *no Brauer–Manin obstruction can exist* on $V_{1,1,1,-1}$. P1 is impossible *a priori*, not merely empirically weak.

This is the cleanest negative result of the session: no Magma computation was needed; the answer was in a single remark in a 2008 paper, surfaced by the retrieval corpus in ~ 10 minutes.

3.5 F1: Peschmann-style fiber filter (sound but unusable at scale)

Peschmann’s recent work on perfect Euler bricks [4, 5] introduces a fiber-quotient framework: fix a parameter, form a genus-3 plane quartic, factor through an elliptic quotient, and prove non-existence by combining rank-zero certification with a Kummer-character analysis on the 2-torsion. This framework adapts directly to our setting. For each sieve candidate (D, C) , set $N := D^4 - C^4$. The fiber $\{a^4 + b^4 = N\}$ is a genus-3 plane quartic; with $W := b^2$ it admits the genus-1 quotient $H_N : W^2 = N - a^4$. The Jacobian of the latter, in standard form, is

$$E_N : Y^2 = X^3 + 27N \cdot X, \quad (3)$$

a curve with $j = 1728$ and CM by $\mathbb{Z}[i]$. Rational integer hits on H_N correspond to rational points (x, y) on E_N such that the preimage on H_N has its y -coordinate a perfect square in $\mathbb{Q}$.

The torsion observation. We ran PARI’s `ellrank` on E_N for each of 11 certifiable known hits and the 4 search-reachable H1 hits. In every case where the rank was certified (14 of 15), the result was $\text{rank}(E_N) = 0$; the remaining 3 cases had bounds $[0, 2]$. In all certified-rank-0 cases the integer solution (a, b) comes from the unique non-identity $\mathbb{Z}/2$ -torsion of E_N . In short:

Observation. Every observed Frye/Lander–Parkin hit corresponds to a 2-torsion orbit on the Jacobian E_N , never to a free Mordell–Weil generator.

Strong form refuted. A natural conjecture extrapolates the observation to all integer pairs: *for every $N = a^4 + b^4$ with $a, b \in \mathbb{Z}$, $\text{rank}(E_N) = 0$* . This is false. Direct `ellrank` on $(a, b) \in [1, 30]^2$ with $\gcd(a, b) = 1$ exhibits 6+ counterexamples with certified $\text{rank}(E_N) \geq 1$:

$$(18, 29), (19, 21), (19, 29), (20, 23), (22, 27), (27, 29). \quad (4)$$

The most likely interpretation of the observation is therefore statistical, not structural: $\text{rank}(E_N) = 0$ is the typical case for random CM curves $Y^2 = X^3 + AX$ (since $L(E_A, 1) \neq 0$ generically and BSD is proven for CM elliptic curves), and the 14 observed Frye hits sample the dominant regime rather than a special subset.

Why F1 fails as a filter. The non-existence criterion is not “ $\text{rank}(E_N) = 0$ ”—that admits torsion-based integer hits, as the regression on $d = 2813001$ shows. The correct criterion is “ $\text{rank}(E_N) = 0$ and H_N is a non-trivial principal homogeneous space for E_N ”. The latter is a 2-Selmer triviality question. PARI’s `hyperellratpoints(f, B)` can probe it directly: for $d = 422481$ ($N \approx 3 \cdot 10^{22}$) a bound $B = 200,000$ suffices and the search finishes in < 1 s; for $d = 2813001$ ($N \approx 4 \cdot 10^{24}$) the required bound is $B > 700,000$ and runtime degrades sharply; for HPC-scale $d \sim 10^{10}$, the required bound is essentially the same as the brute-force decompose loop and runtime is comparable. F1 therefore ships as a diagnostic-only stage in `frye_dc_sieve`—`--pari-filter PATH` writes survivors to a temp file, shells out to a locally-installed PARI/GP, parses verdicts, and reports the certified-rank-0 fraction (typically $\sim 35\%$ at $d \sim 10^{10}$) without rejecting any survivor.

4 Tooling assessment

4.1 Deployment changes made during the session

The tooling discussion below catalogues what the AI *used*; the following list catalogues what the AI *deployed* or *modified* on the user’s running infrastructure during the session. These are

recorded separately because a reader unfamiliar with current AI-assisted research tooling may assume the AI ran read-only analyses alongside the user’s codebase, which is not the case.

- **PARI/GP 2.17.3 installed on the HPC cluster.** Source downloaded over HTTPS, configured with `Configure --prefix=$HOME/packages/pari` against the cluster’s `gcc/13.3.0` module, and built via `make install-bin`. All steps were AI-issued shell commands over SSH. The `gp` binary at `$HOME/packages/pari/bin/gp` (with GMP support) persists across sessions and remains available for any future caller.
- **New C++ integration in `src/frye_dc_sieve.cpp`.** A `--pari-filter` PATH command-line option was added, together with companion options `--pari-filter-script` and `--pari-filter-tmp`; a new `apply_pari_filter()` function with subprocess- plus-file-I/O semantics; five new diagnostic counters (`pari_filtered_in`, `pari_would_reject_rank0`, `pari_rank_positive`, `pari_uncertain`, `pari_total_seconds`); the MPI all-reduce buffer extended from 18 to 23 entries to accommodate them; and an embedded default copy of the F1 PARI script as a literal string so the binary is usable without an external script file. The source was rebuilt with `gcc/13.3.0`, the new `--pari-filter` option verified via `--help`, and the existing 13-hit regression (`./build/frye_dc_sieve $D $D --c-lo $C --c-hi $C --decompose --include-mixed-parit`) confirmed still passing.
- **New Slurm submission script.** `slurm/param_scan_sapphire_dmax15.sbatch` was written from scratch, parameterised for the sapphirerapids partition (112 cores per task, 56 tasks, 4-hour wall, $D_{\max} = 10^{15}$), and submitted via `sbatch`. A separate one-node smoke script was submitted first to confirm the build, module load, and partition policy.
- **Cleanup.** A stale top-level `frye_dc_sieve.cpp` sitting next to `build_cpu.sh` in the project root (which had initially looked like the canonical source and cost roughly an hour of investigation before being identified as unused) was removed. This was a direct deletion, not a rename or archive; the canonical `src/frye_dc_sieve.cpp` is now the single source of truth.
- **Magma input scripts in the project directory.** Roughly a dozen short Magma scripts (`magma_C1.basis.magma`, `magma_torsion.magma`, `magma_sat.magma`, etc.) were written to the user’s local working directory and submitted via `curl` to the public XML calculator endpoint. These are short, single-purpose, and reproducible artifacts.

The user retained the role of final arbiter: each non-trivial deployment (HPC build, Slurm submission, source-level modification, stale-file deletion) was confirmed in dialog before dispatch. Execution was the AI’s; the policy choices and the resource commitments were the user’s.

4.2 Retrieval corpus (math.AG + math.NT)

The single most leveraged service. Three classes of queries were used:

- **Surname lookups** for Bright, Ieronymou, Skorobogatov, Peschmann, Asiryan, Logan, Creutz, Elsenhans, and others. These were the most efficient retrieval class. The Bright surname query in particular surfaced arxiv:1008.0999, in which §2’s parenthetical remark closes P1 for our surface in a single sentence—a result that would have taken several days of speculative cohomology computation to derive from scratch.
- **Equation FTS5 search** for $a^4 + b^4 + c^4$ near d^4 in display math. This returned roughly fifteen papers spanning several adjacent landscapes: the classical Euler problem $A^4 + B^4 = C^4 + D^4$ (math/0505629, 2305.07657, 1610.03430—Tomita’s recreational survey, useful for historical context), Kummer surface theory at the intersection with our K3 (2409.14819, 2402.01223, 1209.0439, 1004.3503), related diophantine work on $a^4 + b^4 + c^4 + d^4 = (a +$

$b + c + d)^4$ (1704.03200), the soluble-local-field side via 2112.10854, and one decidability-of-Diophantine-equations note (2404.01483) that explicitly contrasts Wiles’s negative result on $x^n + y^n = z^n$ ($n > 2$) with Elkies’s positive result on $a^4 + b^4 + c^4 = d^4$, inviting our problem into the “infinitely many solutions known” bucket. None of the returned papers was direct non-existence work on $V_{1,1,1,-1}$, which was itself a useful signal: the brute-force search avenue is not duplicating a closed problem in the corpus.

- **Reading individual papers in full or by section** via `read_paper_section` for the four most decisive retrievals (Bright, IS-Z, Peschmann arxiv:2604.09328, and Logan–McKinnon–van Luijk on density). In each case it was a specific subsection (Bright §2, IS-Z Cor 3.3, Peschmann introduction + discussion, LMvL introduction) that supplied the relevant citation; the full-paper read was the exception.

A recent (Dec 2025) Asiryan preprint on the “Linear Slicing Method for Equal Sums of Like Powers” was surfaced both by the equation search and by direct author lookup; its modular divisibility obstruction (MDO) was quickly assessed and rejected as inapplicable for our $k = 4$ case (the MDO modulus reduces to $M_4 = 2$ —parity only—already implicit in the existing CRT stack). Logan–McKinnon–van Luijk’s 2008 density paper was the decisive citation that ruled out the possibility of a blanket non-existence theorem on $V_{1,1,1,-1}$ (Elkies’s Zariski-/real-analytic-density of $V_{1,1,1,-1}(\mathbb{Q})$).

Coverage gaps were minimal within the math.AG/NT scope; the only friction points were a known FTS5 hyphen tokenisation quirk (“Mordell-Weil” must be quoted as a phrase) and the occasional 50 KB section returned by `read_paper_section`, which can press against context budgets.

4.3 In-process Mathematica

Load-bearing in the structural-diagnosis phase (§2). All five C_i polynomial identities, the smoothness check on ∇F , the $\mathbb{Q}(\zeta_8)$ -twist identification with the Fermat quartic, and the 6×6 intersection-matrix computation that determined $\mathbb{Q}$ -Picard rank exactly 6 were done in Mathematica. Each computation was a few seconds; the chain from “user has five curves of unclear genus” to “catalogue is complete, proceed with H1 group-law work on each $C_i(\mathbb{Q})$ separately” took roughly half an hour of Mathematica work plus the corresponding literature retrieval to confirm the discriminant value -64 and the Kummer-surface identification. The principal value was symbolic manipulation in $\mathbb{Z}[p, q]$ that would have been tedious to set up correctly in any system without a notebook-style REPL: rapid “compute then look at the residual” iterations against parametric identities. Mathematica was not used in the H1, P1, or F1 phases; those moved to Magma and PARI/GP respectively.

4.4 Magma online calculator (programmatically driven)

Magma is a commercial closed-source computer algebra system; the user did not hold a local license. The University of Sydney operates a free public web service¹ that accepts a Magma script in a textarea, runs it under a 120-second wall-clock cap and a 50,000-character input cap, and returns the output formatted for the browser. Normal use is interactive: a human pastes one script, reads the result, edits, resubmits.

For this session every Magma submission was made instead via a scripted HTTP POST to the underlying XML endpoint `magma.maths.usyd.edu.au/xml/calculator.xml`:

```
curl -s --max-time 130 -X POST \
  --data-urlencode "input@magma_C1_basis.magma" \
  https://magma.maths.usyd.edu.au/xml/calculator.xml \
```

¹<https://magma.maths.usyd.edu.au/calc/>


```
-o result.xml
```

The response is an XML document with one `<line>...</line>` element per output line, plus headers reporting wall time and memory. We made roughly fifteen such submissions during the session—for `EllipticCurve(H, P_0)` conversions, `ReducedBasis + HeightPairingMatrix` workflows, `TorsionSubgroup` and `Saturation` probes, and the iterated $[-N..N]^r$ lattice walks of §3.3. Each call returned within seconds for the algebraic-geometry primitives we needed.

Two virtues of the programmatic mode were essential: (i) the same input file is also a reproducible artifact—we keep `magma.C1_basis.magma`, `magma.C2_smallld.magma`, etc. in the project directory and can re-run any analysis without a GUI; and (ii) when one submission outgrows the 120 s cap (we hit this twice—once on a full `Rank` computation at conductor $\sim 10^{20}$, once on $[-10, 10]^3$ enumeration of two curves in a single submission), the script is trivially split into smaller calls without losing state. Both failures were resolved by splitting: the $[-10, 10]^3$ work was distributed across two halves of n_1 , the high conductor was bypassed by switching that computation to PARI/GP (see below).

Programmatically driving a public web service is, to our knowledge, not the standard workflow for users of the Magma calculator; the calculator’s primary interface is the web form. But the XML endpoint is stable, the per-request size is reasonable for substantial algebraic-geometry tasks, and the no-license-required path made it possible to do roughly half the H1 arithmetic without owning Magma. The trade-off is the absence of session state—each submission must recreate the curve, basis, and maps from scratch—and the 120 s and 50,000-character caps which constrain the granularity of work. For the present session both proved manageable: the working `*.magma` scripts in the project directory are short, single-purpose, and reproducible.

4.5 Locally-installed PARI/GP

The decisive tool for F1’s elliptic-rank work. PARI’s `ellrank` for the $Y^2 = X^3 + AX$ family uses a specialised 2-descent that exploits CM by $\mathbb{Z}[i]$, with per-curve cost ~ 30 ms at $d \sim 10^{10}$ ($A \sim 10^{40}$) and ~ 70 ms at $d \sim 10^{12}$. Magma’s online calculator timed out on the same input at $d = 422481$; the gap is purely the specialised algorithm.

PARI was the only tool in the session that successfully ranked E_N at HPC scale. We installed PARI 2.17.3 from source into the user’s HPC `$HOME/packages/pari` (5-minute build, `gcc/13.3.0` module loaded), confirmed it matches the WSL-local install bit-for-bit on a sample, and wired it into `frye_dc_sieve` as `--pari-filter` via file-based subprocess I/O.

4.6 OSCAR/Hecke.jl

Loaded successfully but contributed nothing on the core elliptic-rank question: OSCAR/Hecke surfaces torsion (`torsion.structure`) and several diagnostic invariants, but currently has no `rank(E::EllipticCurve{QQFieldElem})` for elliptic curves over $\mathbb{Q}$. The `mordell_weil_*` primitives are bound to elliptic *surfaces*, a different context. For this session OSCAR was strictly dominated by PARI; we did not investigate whether OSCAR can shell out to PARI internally.

4.7 HPC cluster

The user’s existing build (`frye_dc_sieve`, `param_curve_scan`) ran without modification once module load `gcc/13.3.0` was added (the binaries reference a `GLIBCXX_3.4.29+` runtime not present on the login-node default). We modified `src/frye_dc_sieve.cpp` to add a `--pari-filter/--pari-filter` triple of options. The integration is subprocess-based with a file-based survivor list, embedded default PARI script, and five new MPI-reduced counters (`pari_filtered_in`, `pari_would_reject_rank0`, `pari_rank_positive`, `pari_uncertain`, `pari_total_seconds`). A 56-task Slurm array job on the `sapphirerapids` partition (112 cores per task, 4-hour wall) was submitted at session close on

the parametric scanner with $d_{\max}$ raised to 10^{15} and $Q_{\max}$ widened by $1.33\times$. All 56 tasks completed within 2.3–2.5 hours each, well inside the wall. The job produced 19 hit lines reducing to **15 distinct primitive solutions**: all 13 catalogued hits were recovered (regression of the prior search) and both H1 search-reachable hits inside the (p, q) window were independently found— $d = 2051\,764\,828\,361$ on C_2 via $v = 1026427/1526709$ and $d = 29\,999\,857\,938\,609$ on C_1 via $v = 5207881/1383765$. No surprise hits beyond H1’s predictions appeared in this window, which is the clean “no false positives” outcome. The two H1 hits at $d = 4.56 \cdot 10^{13}$ and $d = 5.74 \cdot 10^{14}$ lie outside this window’s (p, q) box and were correctly absent from this run.

5 Discussion

Borrowing from the literature is the point. The mathematical community’s accumulated results on diagonal quartic surfaces, Brauer groups of Kummer surfaces, and the arithmetic of elliptic curves with CM by $\mathbb{Z}[i]$ are exactly what enabled this session to make sharp claims in hours rather than weeks. P1 looked tractable at first reading—the IS-Z framework computes Br of any diagonal quartic up to a known bound, and our $|H| = 4$ case is well within Magma’s reach. Implementing the cohomology computation in Magma or OSCAR would have been a multi-day project. Bright’s remark in [2] short-circuited the entire calculation: $\text{Br}_1(X) = \text{Br}(\mathbb{Q})$ for our family follows from his thesis tables in the explicit family $(1, c_1, c_2, -c_1^2 c_2^2)$. Surname-keyed retrieval surfaced this remark in a single tool call. Citing a published theorem rather than re-deriving it is a feature, not a shortcut: the result is more reliable, has been refereed, and explicitly invites future readers to consult the cited literature for the cohomology details. Similar borrowing happened for the Schütt–Shioda–van Luijk discriminant -64 that anchored §2, for the density theorem of Elkies that ruled out blanket non-existence, and for Peschmann’s recent fibre-quotient framework that gave the F1 reduction *for free*.

Where the next collaborator could go further. The most computationally interesting open question—can F1 be made to reject survivors correctly?—reduces explicitly to a 2-Selmer triviality test on H_N for each (D, C) . This is a well-defined problem within the existing 2-descent literature ([4], the Bruin–Stoll line of work on hyperelliptic 2-descent, and Magma’s `TwoSelmerGroup` intrinsic). PARI’s `hyperellratpoints` solves the question for small enough height bound, but the required bound grows linearly with $N^{1/4}$, matching the cost of the brute-force decompose loop that F1 was intended to short-circuit. Magma’s `TwoSelmerGroup` on the genus-1 Jacobian is in principle the right tool but times out on the online calculator at $d \sim 10^{10}$. A collaborator with a local Magma license, or someone who could implement specialised PARI-level 2-descent exploiting the CM structure of E_N (drawing on Damerell–Coates–Wiles for the L -value side and on Peschmann’s Kummer-character analysis on the torsion side), would close the gap. We invite this work explicitly; the F1 diagnostic stage in `frye_dc_sieve` is already wired to consume any verdict file a stronger backend produces.

H1 in retrospect. The largest single arithmetic deliverable was the H1 hit catalogue (Table 2). Two structural facts surfaced incidentally. First, the Mordell–Weil ranks of $C_0, \dots, C_4$ are not the rank 1 noted in the early session tracking but $\{0, 3, 3, 2, 2\}$; the user’s catalogue of 12 known C_1 seeds is exhausted at $[-2, 2]^3$ basis coefficients with a single torsion shift. Second, the H1 hit $d = 5.74 \cdot 10^{14}$ is the first known hit whose smallest (p, q) representative $(1, 251, 197, 642, 314, 528, 967)$ exceeds the rectangular window of any practical (p, q) scanner; group-law iteration is genuinely necessary, not merely convenient, to enumerate this point.

The torsion observation, refined. At session start it was tempting to read the 11-of-13 rank-zero pattern as a structural theorem. The 30-second test on $(a, b) \in [1, 30]^2$ refuted the strong form: $\text{rank}(E_{a^4+b^4})$ takes positive values often enough to make the all-rank-zero

hypothesis untenable in general. The 11-of-13 plus 14-of-17 sample across known and H1 hits, combined with the unanimous absence of certified rank ≥ 1 in either set, remains noteworthy and possibly correlated with the simultaneous-solvability constraint on (c, d) —but it does not yield a search shortcut.

Localising the open work is a deliverable. Two of the five sub-projects closed by hand-off rather than by internal computation: P1 closes via the Bright theorem (no further work needed; the result is published), and F1 closes to the 2-Selmer triviality question on H_N (a well-defined problem with an identifiable expert audience). In both cases the closure took a single decisive step. Negative-by-handoff results are cheap when the right reference or the right reduction is identified early; the session’s three rounds of literature scanning (math.AG cluster ranking, math.NT cluster ranking, equation search, six author lookups) shaped both closures. What remains in the open-questions list is not residual failure but explicit work-units the next collaborator can pick up.

6 Open questions and invitations for collaboration

The session closed four queued sub-projects and discovered one was already implemented; what remains are four well-localised questions, each with an identifiable expert audience that could carry the work forward. These are not gaps in the present writeup—they are deliberately exposed interfaces.

1. **2-Selmer triviality on H_N at HPC scale** (for the Diophantine-geometry / explicit-descent community). Making F1 a real rejecting filter reduces explicitly to deciding, for each candidate (D, C) , whether the principal homogeneous space $H_N : y^2 = N - x^4$ is trivial as an element of $\text{Sel}^{(2)}(E_N/\mathbb{Q})$. Magma’s `TwoSelmerGroup` handles this in principle; the online calculator does not have the wall-clock budget for it at $d \sim 10^{10}$, and a local Magma license or a specialised PARI-level implementation in the Damerell–Coates–Wiles framework—exploiting the CM by $\mathbb{Z}[i]$ of the $j = 1728$ family—would close the question. The `frye_dc_sieve` integration is already file-based: whatever backend produces $(D, C, \text{has_Q_point})$ verdicts can be plugged in without further C++ work.
2. **Why does every observed Frye hit have rank(E_N) = 0?** (for the analytic-rank / Chowla–Selberg community). The 11-of-13-then-14-of-17 empirical pattern across known and H1 hits is suggestive but does not survive extrapolation to general $N = a^4 + b^4$ (§3.5). Two routes to clarification: a heuristic count via random- L -value statistics in the CM family, or an effective $L(E_N, 1) \neq 0$ argument exploiting the multiplicative structure $N = (D - C)(D + C)(D^2 + C^2)$ over $\mathbb{Z}[i]$. Either would be of independent interest in the arithmetic of Mordell curves.
3. **Does the hit $d = 5.74 \cdot 10^{14}$ have a smaller-coordinate (p, q) representative?** (for anyone interested in the geometry of $\text{MW}(C_1)$). The two representatives produced by our group-law enumeration both have $|p|, q \gtrsim 10^8$, exceeding by orders of magnitude the rectangular scan windows of any past or feasible parametric search. Either this is a structural feature of the corresponding $\text{MW}(C_1)$ orbit (a claim worth proving), or there is a smaller-coordinate representative that the lattice walk in §3.3 did not visit. An explicit covolume / 2-isogeny descent on C_1 could decide this.
4. **C++ group-law iteration over the H1 basis** (for the HPC / GMP-engineering side). Retrofitting `param_curve_scan` with explicit $n_1B_1 + n_2B_2 + n_3B_3 + tT$ enumeration in GMP arithmetic, replacing the rectangular (p, q) scan, would produce all primitive hits up to any chosen d_{max} on the C_0 – C_4 branch without the Magma 120 s submission bottleneck.

Engineering is well-defined: the inverse map from $(X, Y) \in E_m$ back to $(p, q) \in H$ is explicit from the Magma session output and can be transcribed into GMP-aware C++. Estimated 2–3 days of focused work.

Each item names a concrete community of practitioners who would benefit from—and could move forward—the work the session has left in their lane.

7 Acknowledgments

This document was drafted by Claude (Anthropic), model Opus 4.7, during the same single session it describes. The user provided the problem, the HPC cluster account and the prior infrastructure (`frye_dc_sieve`, `param_curve_scan`, prior Slurm runs, the session-tracking PROGRESS.md notes that emerged during the work), supervised tool authorisation decisions (notably the Magma online-calculator submissions and the HPC `sbatch` dispatch), and acted as final arbiter on direction. All arithmetic in Tables 1–2 was verified independently in Python and/or Magma; the assertion $\text{Br}_1(V_{1,1,1,-1}) = \text{Br}(\mathbb{Q})$ is a quotation from [2], not an independent computation.

References

- [1] N. Elkies, *On $A^4 + B^4 + C^4 = D^4$* , Math. Comp. **51** (1988), no. 184, 825–835.
(Density of $V_{1,1,1,-1}(\mathbb{Q})$ as cited in Logan–McKinnon–van Luijk, arXiv:0812.4779.)
- [2] M. Bright, *The Brauer–Manin obstruction on a general diagonal quartic surface*, arXiv:1008.0999.
- [3] E. Ieronymou, A. N. Skorobogatov, Yu. G. Zarhin, *On the Brauer group of diagonal quartic surfaces*, arXiv:0912.2865.
- [4] R. Peschmann, *Quartic reductions and elliptic obstructions for perfect Euler bricks*, arXiv:2604.09328.
- [5] R. Peschmann, *A torsion-intersection proof of perfect-cuboid nonexistence on 1,072 explicit master-tuple fibers*, arXiv:2604.28072.
- [6] V. Asiryan, *The Linear Slicing Method for Equal Sums of Like Powers: Modular and Geometric Constraints*, arXiv:2512.00551 (Dec 2025).
- [7] A. Logan, D. McKinnon, R. van Luijk, *Density of rational points on diagonal quartic surfaces*, arXiv:0812.4779.
- [8] User’s working C++ search codebase (`param_curve_scan`, `frye_dc_sieve`), prior Slurm output files documenting the 13-hit catalogue, and session-tracking PROGRESS.md notes (all project-local, 2026-05).